THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 1: **Real Numbers** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The prime factorisation of 156 is: 1

(a) $2^2 \times 3 \times 13$ **(b)** $2 \times 3^2 \times 13$ **(c)** $2^2 \times 3^2 \times 13$ **(d)** $2 \times 3 \times 13$

2. Using prime factorisation, HCF$(96, 404)$ is: 1

(a) 2 **(b)** 4 **(c)** 8 **(d)** 12

3. If HCF$(a, b) = 12$ and $a \times b = 1800$, then LCM(a, b) is: 1

(a) 150 **(b)** 1800 **(c)** 180 **(d)** 120

4. A student states: "The product of two irrational numbers is always irrational." The correct refutation is: 1

(a) The statement is correct **(b)** False, since $\sqrt{2} \times \sqrt{2} = 2$ is rational

(c) True only for square roots **(d)** Cannot be determined

5. Which of the following is an irrational number? 1

(a) $\sqrt{16}$ **(b)** 0.25 **(c)** $3 + \sqrt{5}$ **(d)** $\frac{22}{7}$

6. The number 4^n, where n is a natural number, ends with the digit 0 for: 1

(a) $n = 5$ **(b)** $n = 10$

(c) every even n **(d)** no natural number n

7. The LCM of the smallest prime number and the smallest composite number is: 1

(a) 2 **(b)** 4 **(c)** 6 **(d)** 8

8. Two students find the HCF and LCM of 26 and 91. Student A gets HCF $= 13$, LCM $= 182$; Student B gets HCF $= 13$, LCM $= 2366$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

9. If p is a prime number, then $\sqrt{p}$ is: 1

(a) rational **(b)** irrational **(c)** an integer **(d)** a natural number

10. The HCF of two consecutive natural numbers is: 1

(a) 0 **(b)** 1

(c) the smaller number **(d)** their product

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Express 3825 as a product of its prime factors. 2

12. Find the HCF and LCM of 12, 15 and 21 by the prime factorisation method. 2

13. Find the HCF and LCM of 17, 23 and 29 by prime factorisation. What do you observe? 2

14. Find the HCF and LCM of 336 and 54, and verify that HCF $\times$ LCM $=$ product of the numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sqrt{3}$ is an irrational number. 3

16. There is a circular path around a sports field. Sonia takes 18 minutes to drive one round, while Ravi takes 12 minutes. If they start at the same point at the same time in the same direction, after how many minutes will they meet again at the starting point? 3

17. Prove that $3 + 2\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational. 3

OR

Show that $11 \times 13 \times 17 + 17$ and $3 \times 5 \times 7 \times 11 + 11$ are composite numbers. 3

18. An army contingent of 616 members is to march behind an army band of 32 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march? 3

OR

Three bells toll at intervals of 9, 12 and 15 minutes respectively. If they toll together at 8:00 a.m., at what time will they next toll together? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Find the HCF and LCM of 510 and 92 by prime factorisation and verify that HCF $\times$ LCM $=$ product of the two numbers. 5

(ii) A student claims that two numbers can have HCF 18 and LCM 380. Justify or refute the claim.

OR

Prove that $\sqrt{2}$ is irrational. Using this result, show that $5 - 3\sqrt{2}$ is also irrational. 5

20. **Case study:** A sweet seller has 420 kaju barfis and 130 badam barfis. She wants to stack them so that each stack has the same number of barfis and contains only one type, using the least area of the tray. 5

(i) What is the maximum number of barfis in each stack?

(ii) How many stacks of each type will there be?

(iii) If she also has 260 pista barfis, will the number of barfis per stack change? Justify.

OR

Case study: Two tankers contain 850 litres and 680 litres of kerosene. 5

(i) Write the prime factorisation of 850 and 680.

(ii) Find the maximum capacity of a container that can measure the kerosene of both tankers an exact number of times.

(iii) How many times will each tanker be measured? Verify HCF $\times$ LCM $= 850 \times 680$.

END OF QUESTION PAPER — **Total: 40 Marks**